



### Exercise 1

Consider

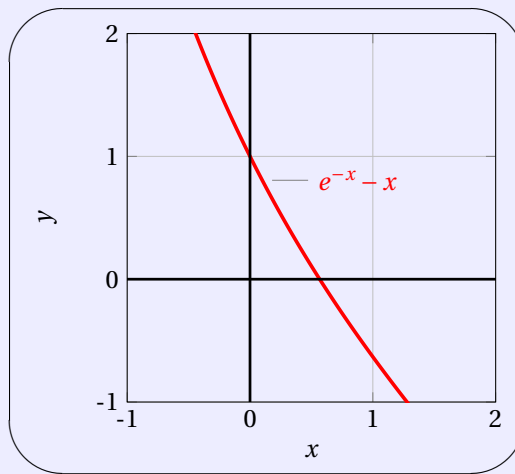
$$f(x) = e^{-x} - x$$

and verify whether Newton's method can be applied on the interval  $[0, 1]$ . If so, find an estimate of  $x \in [0, 1]$  such that:

$$e^{-x} - x = 0$$

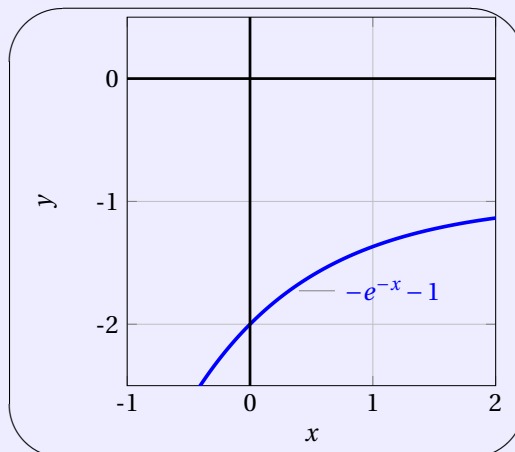
We consider the function:

$$f(x) = e^{-x} - x, \quad f'(x) = -e^{-x} - 1, \quad f''(x) = e^{-x}$$



We have:

$$f(0) = 1 > 0 \quad \text{and} \quad f(1) < 0$$



We consider the interval  $[0, 1]$ ; for  $x \in [0, 1]$  we have:

$$f'(x) < 0 \quad \text{and} \quad f''(x) > 0$$

Hence on the interval  $[0, 1]$  the function  $f(x) = e^{-x} - x$  satisfies the hypotheses of the theorem.

We are in the case of a function that is strictly decreasing on  $[0, 1]$ , since  $f'(x) < 0, x \in [0, 1]$ , and convex on  $[0, 1]$ , since  $f''(x) > 0, x \in [0, 1]$ . The endpoints of the chosen interval are:  $a = 0$  and  $b = 1$ .

Hypothesis 3 holds, that is  $f(0) \cdot f''(0) > 0$ , and the sequence becomes:

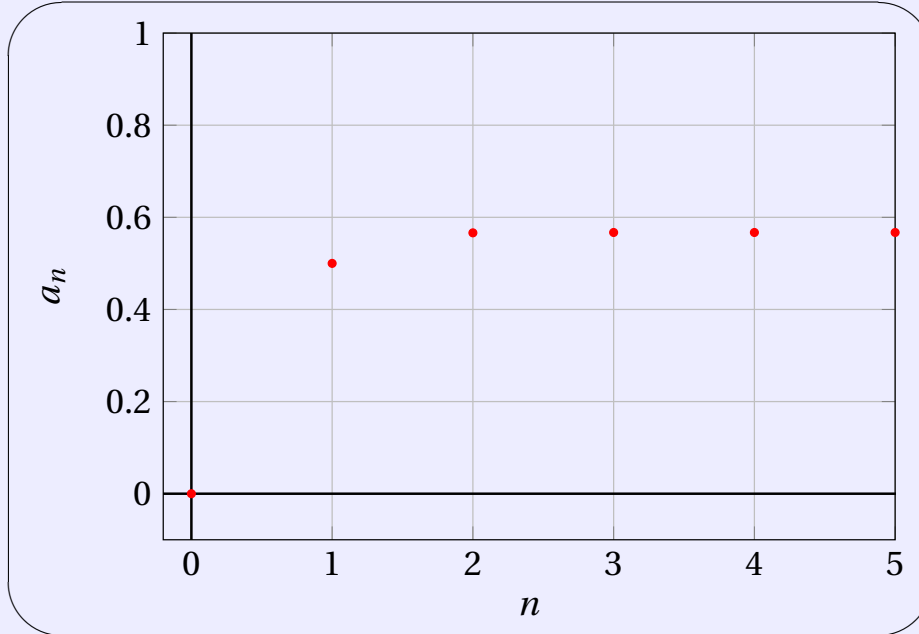
$$x_0 = 0, \quad x_{n+1} = x_n - \frac{e^{-x_n} - x_n}{-e^{-x_n} - 1} = \frac{x_n + 1}{1 + e^{x_n}} \text{ with } n \in \mathbb{N}$$

Since

$$\begin{aligned} x_n - \frac{e^{-x_n} - x_n}{-e^{-x_n} - 1} &= x_n + \frac{e^{-x_n} - x_n}{e^{-x_n} + 1} = x_n + \frac{e^{-x_n}}{e^{-x_n} + 1} \frac{1 - e^{x_n}}{1 + e^{x_n}} \\ &= \frac{x_n(1 + e^{x_n}) + 1 - e^{x_n}}{1 + e^{x_n}} = \frac{x_n + 1}{1 + e^{x_n}} \end{aligned}$$

we therefore have:

$$x_0 = 0, \quad x_1 = 0.5, \quad x_2 = 0.5663\dots, \quad x_3 = 0.5671\dots, \quad x_4 = 0.5671\dots, \quad x_5 = 0.5671\dots$$



At the first iteration the tangent line is:

$$y = 1 - 2x \text{ and } x_1 = 0 - \frac{1}{-2} = \frac{1}{2}$$

